# Generating ideas

Name: Unit:

1 I write down one idea that works.

2 I write down several ideas that work in different ways, not small changes of the same idea.

3 I sort my ideas into kinds and add one from a kind I had missed.

4 I pick the idea that fits best when one condition changes, explain why, and say how sure I am.

The ladder is for students' own self-check. It is never graded. Circle the step your answer reached.

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# Evaluating and choosing

Name: Unit:

1 I make a choice or a judgement and give one reason for it.

2 I give several reasons, each backed by a fact or a rule I can check.

3 I weigh my reasons against each other and against the other side, and show which one decides it.

4 I test my judgement on a new case or a new piece of evidence, and say how sure I am.

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# Comparing and transferring

Name: Unit:

1 I name one thing the two cases share.

2 I name several things that match and at least one place where the cases differ.

3 I explain the rule that makes the cases match, and where the comparison breaks.

4 I use the rule on a new case, and say how sure I am that it fits.

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# Explaining and reasoning from basics

Name: Unit:

1 I give one explanation or one fact I know for certain.

2 I give several explanations or steps, and say what each one rests on.

3 I link my steps into one line of reasoning, or show which explanation fits the evidence best.

4 I use my reasoning on a new case or a changed number, say where it stops working, and say how sure I am.

The ladder is for students' own self-check. It is never graded. Circle the step your answer reached.

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# Causes and systems

Name: Unit:

1 I name one cause and one effect.

2 I build a chain: A leads to B leads to C.

3 I find a loop: C affects A again, or a delay or a side effect.

4 I use the loop to predict what happens if one part changes, and say how sure I am.

The ladder is for students' own self-check. It is never graded. Circle the step your answer reached.

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# Regulating your own thinking

Name: Unit:

1 I name one thing I did while I worked.

2 I describe several steps in how I worked, and one moment that shaped how I worked, such as changing my approach or getting stuck.

3 I explain what made me change my approach or moved me on, and why it helped.

4 I say what I would change next time and why, and how sure I am that it would help.

Students rate the ladder on the Think back part at the end of each task, not on the task answer.

The ladder is for students' own self-check. It is never graded. Circle the step your answer reached.

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